



Thermal and power consumption model of an electrically refrigerated trailer

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ARTICLE INFO

Keywords:

Transport refrigeration
Refrigerated vehicle
Lumped capacitance
Thermal model
Power model

ABSTRACT

In the face of escalating climate change concerns, there is a need for reducing CO₂ emissions in all economic sectors. The transportation sector, particularly freight transportation via trucks and especially refrigerated trailers, is a substantial contributor to these emissions. An energy management system can be used to optimize the energy consumption of electrified refrigerated trailers, for which a model of an electrically refrigerated trailer is needed. This paper aims to develop and validate a thermal model of an electrified trailer. The model is created using over 40 h of test drives across a broad spectrum of internal temperatures, which range between -23 °C and +18 °C, as well as ambient temperatures, which range between 1 °C and 36 °C. The data was collected from test drives conducted in Spain, Sweden, and Germany under varying temperature set-points and loads, ensuring the model's accuracy across diverse operating conditions. Additionally, for the first time, the Trailer Refrigeration Unit's (TRU) electric consumption is modeled in detail over complete drive cycles. The paper also outlines a method for identifying the parameters of the trailer and goods across all runs. This comprehensive approach ensures a robust and accurate model, with an RMSE of below 0.7 °C for the open-loop temperature modeling and a total error of below 1.5% for the energy consumption in a closed-loop simulation.

1. Introduction

As the impacts of climate change become increasingly evident, the urgency to reduce CO₂ emissions across all sectors has intensified. In 2020, the transportation sector emerged as the second-largest contributor to global CO₂ emissions, exceeded only by the heat and energy sector. The total emissions of the transportation sector surpassed seven billion tons of CO₂ annually (Ritchie et al., 2020).

Passenger vehicles, including cars and buses, account for approximately 45% of the sector's emissions. Freight transportation, including trucks and lorries, contributes about 30% to the sector's total CO₂ emissions (Ritchie, 2020).

The transportation of refrigerated goods is particularly carbon-intensive, because the Transport Refrigeration Unit (TRU) is typically powered by a diesel generator. Already in 2010, the estimated global fleet of refrigerated vehicles stood at about four million, with 30% being refrigerated trailers, 30% large trucks, and 40% small trucks and vans (United Nations Environment Programme (UNEP), 2010). Given an average of 1500 operating hours per trailer and year and a consumption of 2.5 liters of diesel per operating hour (Otten et al., 2015), refrigerated trailers alone would emit 13 million tons of CO₂ annually

To reduce and evaluate the emission of Greenhouse Gases (GHG), extensive research has been done to create new motorizations for as well as model and improve TRUs and refrigerated trailers. These motorizations include hydrogen-powered trailers (Segura et al., 2023), battery electric, or hybrid trailers (Tobias et al., 2024; Maiorino et al., 2024a). Other approaches include the addition of Phase Change Materials (PCMs) to reduce the temperature fluctuations and overall power consumption (Ben Taher et al., 2022b,a).

One of the most commonly used thermal trailer models was developed by Artuso et al. (2019), in which a small insulated truck body was accurately modeled as lumped heat capacities and validated using a stationary test station. However, because the model consists only of the truck body, it does not incorporate a model of the TRU and its power demand.

In a recent study, the power demand of a diesel-powered TRU was modeled, and the potential CO₂ savings when switching to a battery-powered TRU were estimated to be around 10 million tons annually (Fabris et al., 2022). Furthermore, this and other studies modeled the heat infiltration load during door openings (Fabris et al., 2022; Lafaye de Micheaux et al., 2015; Jeong et al., 2022; Ben Taher et al., 2021), leading to more accurate results during delivery drives. One

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Nomenclature

C	Lumped heat capacity (J K^{-1})
c	Acceleration Coefficient
K	Heat Transfer Coefficient (W K^{-1})
k	Timestep
m	Number of drive-cycles
n	Number of time-steps
P	Power (kW)
Q	Heatflow (kW)
r	Random number between 0 and 1
T	Temperature ($^{\circ}\text{C}$)
v	Particle velocity
w	Inertia of particle
x	Particle position
gb	Global best position over all particles
pb	Personal best position of a particle

Abbreviations

COP	Coefficient of Performance
GHG	Greenhouse Gas
MPC	Model Predictive Control
PLM	Piecewise Linear Model
PSO	Particle Swarm Optimization
RMSE	Root Mean Square Error
TRU	Transport Refrigeration Unit

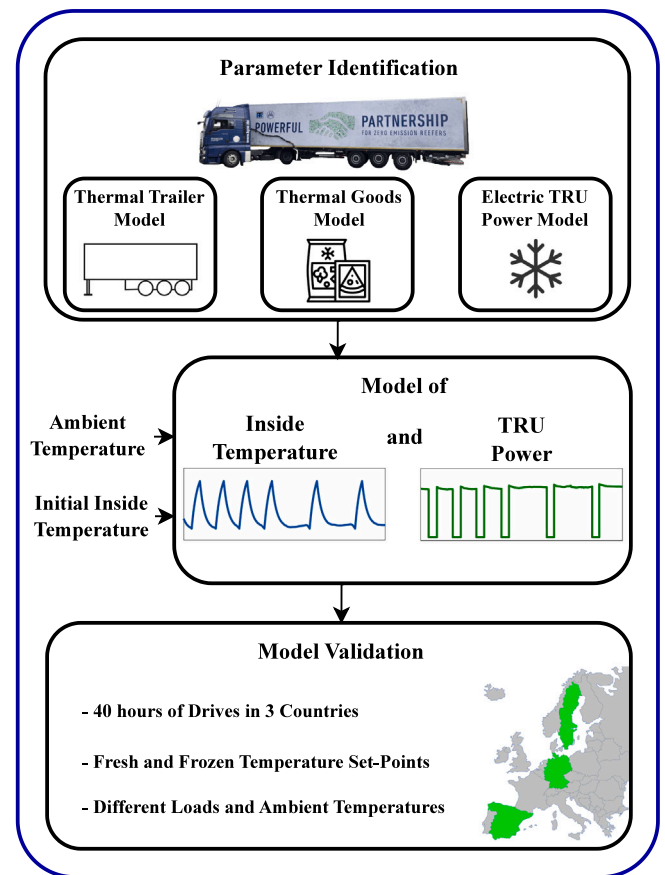


Fig. 1. Overview of the three sub-models: The thermal trailer model, the thermal goods model, and the model of the electrically powered TRU, which are used to model the inside temperature and power consumption of the refrigerated trailer.

of the most used approaches is to simulate the inside air using Computational Fluid Dynamics (CFD). These CFD approaches can model the inside airflow inside the trailer and the air exchange during door openings (Ben Taher et al., 2022a). While these models are accurate, they are more computationally expensive than lumped heat capacity models.

Sørensen et al. developed a model of a reefer container for a Model Predictive Control (MPC), which reduces the energy consumption of a diesel-powered TRU by up to 21% (Sørensen et al., 2015a,b).

While most research focuses on modeling either only the internal temperature of the trailer or a trailer with diesel-powered TRUs, Kulkarni et al. have expanded this scope. Their work includes a diesel-powered TRU model and a Fuel Cell Hybrid model. Based on a lumped capacity approach, these models have been validated using an empty trailer on limited data (Kulkarni and Dwyer, 2007).

Other approaches that were explored in simulation in the past include combining a battery with solar panels (Meneghetti et al., 2021) to power the TRU or combining a battery with a generator axle for breaking energy recovery (Maiorino et al., 2024b; Zeller et al., 2023; Tobias et al., 2024; Bank et al., 2024), showing the potential of using multiple energy sources as well as the need for an energy management system.

The optimization of energy flows for vehicle propulsion in vehicles with more than one type of energy storage has been extensively researched (Lü et al., 2022) and has shown saving potential whenever a vehicle has multiple energy storages. Similarly, multiple sources have suggested or demonstrated that an MPC-based control strategy helps in improving the efficiency of refrigerated transport, especially when using multiple energy sources (Sørensen et al., 2015b; Kulkarni and Dwyer, 2007; Meneghetti et al., 2021; Bank et al., 2024; Li et al., 2010).

An accurate model of the trailer and its goods is essential to develop such an MPC-based control strategy (Sørensen et al., 2015a,b). Most existing literature either does not consider the load or models it as a single heat capacity (Artuso et al., 2019; Fabris et al., 2022; Kulkarni and Dwyer, 2007), which might lead to larger model errors. Furthermore, to

the best of our knowledge, only validated models of diesel TRUs exist. Their validation is usually limited to a single temperature set-point and a narrow range of ambient temperatures. The few models that exist of electric TRUs are not validated and make idealized assumptions of the COP (Kulkarni and Dwyer, 2007; Meneghetti et al., 2021; Maiorino et al., 2024a). Finally, commonly used models like (Artuso et al., 2019) require the solar influx as an input parameter, which is not easily estimateable, because it is highly dependent on many different influences, like the current weather and the diffusion and reflection of the light by the surroundings, making it a non-robust input for a model in an MPC.

Therefore, this paper's objective is to develop a thermal model of the trailer and goods, as well as the first validated power consumption model of an electric TRU, together with a parameter identification strategy. The models should be accurate while requiring a small parameter set, and the input of the final model should only consist of the ambient temperature because it can be estimated over long time horizons and the initial inside temperature as the start point. The developed models will be validated at different temperature set-points with varying loads and ambient temperatures to demonstrate the model's accuracy even under a wide range of operating conditions, resulting in the first validated model that is accurate under a wide range of ambient and internal temperatures as well as the first validated power consumption model of an electric TRU (see Fig. 1).

The remaining paper is structured as follows: Section 2 describes the proposed model's fundamentals, and Section 3 details the parameter identification process. Section 4 gives the test drives undertaken. Section 5 shows the results of the open-loop temperature modeling,

Section 6 the results of the TRU power consumption model, and Section 7 the results of the complete closed-loop model. Finally, Section 8 concludes the paper.

2. Thermal trailer and TRU power consumption model

The thermal model of the trailer is based on a lumped heat capacity model similar to previous approaches (Artuso et al., 2019; Kulkarni and Dwyer, 2007). However, the model differs in crucial points; for example, our model does not require heat influx through solar radiation or air leakage through the doors as input parameters, because they would not be easily estimateable under real-world conditions. The heating caused by solar radiation and air leakage is thus modeled implicitly as part of the heat resistance of the trailer insulation. The trailer is modeled as a single lumped heat capacity with a thermal heat transfer coefficient of K_1 to the ambient air, which models the trailer's insulation, and a second coefficient K_2 that models the heat transfer between the trailer and the inside air, which then exchanges heat with the goods.

To model the cooling power of the TRU, the TRU power consumption is multiplied by a Coefficient of Performance (COP) value. Because the COP is dependent on the inside and ambient temperature and especially the difference between them (Ommen et al., 2019), we identify a Piecewise Linear Model (PLM) for the COP, with the inside and ambient temperature as input values. The PLM consists of a 2D grid of support points, in between which the output is a linear interpolation between the nearest identified parameters. The required PLM size will be discussed in Section 5.1.

The interpolation can be described by

$$f(x, y) = \frac{1}{(x_2 - x_1)(y_2 - y_1)} \cdot [x_2 - x] \cdot [y - y_1] \cdot \begin{bmatrix} f(z_{11}) & f(z_{12}) \\ f(z_{21}) & f(z_{22}) \end{bmatrix} \cdot \begin{bmatrix} y_2 - y \\ y - y_1 \end{bmatrix} \quad (1)$$

where x and y are T_{Amb} and T_{Air} and are lying in between the support points so that $x_1 < x < x_2$ and $y_1 < y < y_2$. Finally, $f(z)$ are the identified COP parameters at the specific ambient and inside temperatures. The PLM size describes the number of support points used for T_{Amb} and T_{Air} so that for a 3×3 PLM, a total of nine COP values were identified. The parameter identification process is described in detail in Section 3.

The TRU power consumption itself was observed to depend on the inner and ambient temperature. Similarly to the COP, a PLM was identified to model the TRU power, the results of which are given in Section 6, in contrast to the COP the TRU parameters can be identified directly, since the power consumption was measured.

Thus, the heat flow caused by the TRU is calculated as

$$Q_{\text{TRU}} = P_{\text{TRU}} \cdot \text{COP}. \quad (2)$$

P_{TRU} and COP represent the power consumption and COP values derived from the PLMs depending on the current inside and ambient temperature. An overview of the complete trailer model is given in Fig. 2.

The heat flow caused by the temperature difference between the inside and ambient temperature is calculated as

$$Q_{\text{Amb}} = (T_{\text{Amb}} - T_{\text{Air}})K_T, \quad (3)$$

with K_T being the heat transfer coefficient, representing the trailer's insulation and implicitly incorporating the air leakage and solar influx.

As detailed previously, the goods are commonly modeled as a single heat capacity, which models the cargo as having a uniform temperature. To improve the model quality, the goods can be modeled as a series of heat capacities and resistances, as shown in Fig. 3. A trade-off has to be made between model complexity and accuracy. To determine the number of needed heat capacities, three different models for the goods were evaluated, and the results are shown in Section 5.2.

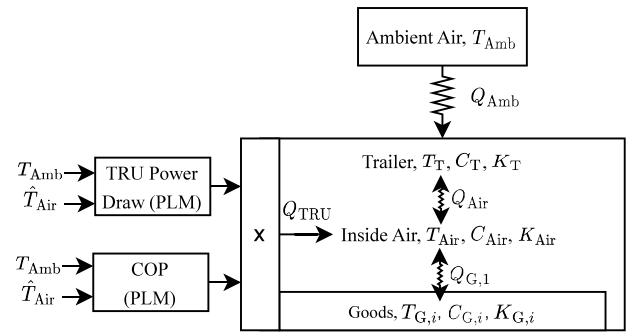


Fig. 2. Overview of the different heat flows in the model. The heat flow caused by the TRU is modeled by two PLMs, which model the current TRU power draw and COP depending on the temperature. Furthermore, the trailer exchanges heat with the inside air, which interacts with the loaded goods.

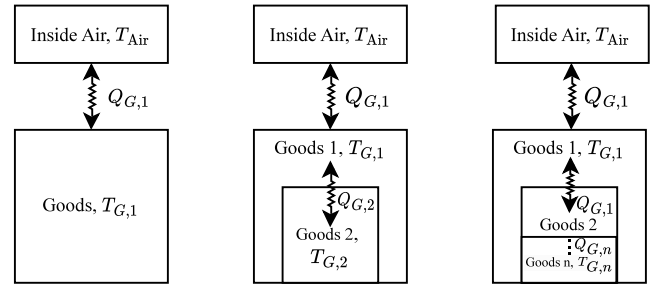


Fig. 3. Thermal goods model. The transported goods can be modeled as a series of n heat capacities and resistances. For $n = 1$ the temperature of the goods is assumed to be homogeneous. At the same time, an increased number of capacities allows for a more detailed description of the goods; it requires a higher model complexity and more parameters to be identified.

The heat flow between the multiple heat capacities is modeled analogously to the ambient heat flow with heat transfer coefficients $K_{G,i}$

$$Q_{G,i} = (T_{G,i} - T_{G,i-1})K_{G,i}, \quad (4)$$

where $Q_{G,1}$ is the heat flow between the outermost layer of the goods and the trailer and all other $Q_{G,i}$ represent the heat flow between the different modeled layers of the transported goods.

The temperature of the inside air in the next time step k can then be calculated as

$$T_{k+1} = T_k + (Q_{\text{Air}} + Q_{G,1} - Q_{\text{TRU}})/C_{\text{Air}} \quad (5)$$

with C_{Air} being the lumped thermal capacity of the inside air. The temperature updates for the different heat capacities of the goods, as well as the trailer itself, can be calculated analogously. This model assumes that each of the modeled thermal capacities has a homogeneous temperature and a homogenous heat transfer to the surrounding thermal capacities. This might impact the modeling of the air, as the inside temperature distribution is not modeled, and the different heat transfer rates are caused by differences in air flow in different parts of the trailer. The number of Floating Point Operations (FLOPS) required to run one inference step of the model is given in Table 1. The computational cost increases by 7 FLOPS for each additional heat capacity that the goods are modeled with, which is a 12% increase.

Other effects, like the inside airflow or temperature distribution, are not modeled, because they vary with the load distribution (So et al., 2021) and thus do not allow for a generalized model. Furthermore, the heat of respiration, emitted as sugars in the transported goods break down (Becker et al., 1996), is not part of the goods model. However, the model could be expanded by adding a heat source to the goods model. Finally, environmental factors like the humidity, road conditions and

Table 1

Computational cost of one inference step of the model depending on the number of heat capacities the goods are modeled with.

Nr. of heat capacities of goods	1	2	3
Computational cost in FLOPS	59	66	73
Relative computational cost	100%	112%	124%

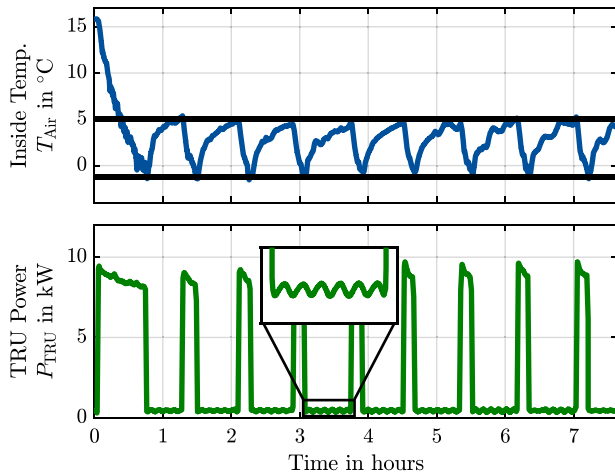


Fig. 4. An example of a cycle sentry drive. The temperature is held between two temperature bands. The TRU power consumption depends on the inside and ambient temperature. In contrast to traditional diesel-powered TRUs the fan can be kept on without the evaporator, leading to a constant power draw, even when cooling is off.

wind-speed are not part of the model, as these parameters might be difficult to obtain during regular operation. However, including these parameters might further improve the accuracy of the models.

As in the existing thermal models, the identified parameters are trailer-specific and change with time as the insulation of a refrigerated trailer deteriorates (Siam et al., 2022), requiring regular re-identification. The good's parameters change with the load. However, the proposed parameter identification process allows for the usage of data collected during normal driving operations without the need for prior knowledge of the trailer's or good's parameters and the ability to use multiple drives with varying loads.

Depending on the loaded goods and their temperature requirements, the trailer can be operated in two different modes: the cycle sentry and continuous cooling mode, which will be explained in detail below.

2.1. Cycle sentry cooling

The cycle sentry cooling mode is commonly used to transport frozen goods that are not easily perishable and do not need a constant temperature and thus can be transported within a temperature range. The cycle sentry mode uses a bang–bang controller, keeping the inside temperature between two temperature bands. When the temperature exceeds the upper limit, the TRU is powered on until the inside temperature reaches the lower set temperature level. An example of a test drive using the cycle sentry mode is shown in Fig. 4.

In contrast to diesel-powered TRUs, where usually the complete diesel engine shuts off, including the fan that circulates the air inside the trailer during the off phase of the cycle sentry mode, in an electric TRU, the fan can stay powered on at all times, ensuring a more even temperature throughout the whole trailer.

In the TRU model, the fan power is estimated at a constant load of 450 W, because it is activated at all times, and 450 W is its median power consumption.

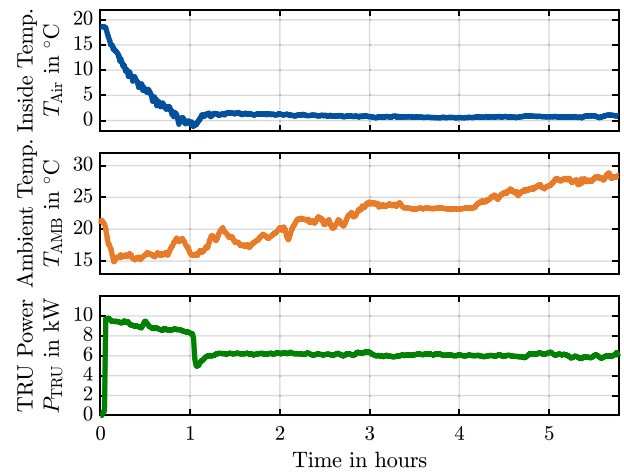


Fig. 5. Example test drive using the continuous cooling mode. The transition between the pull-down and the continuous cooling phase can be seen around the 1-h mark, at which the power consumption drops to a constant value and the inside temperature is held at a constant level.

2.2. Continuous cooling

The second standard operating mode of TRUs is the continuous mode for highly perishable goods, which need to be cooled to a constant temperature.

It can be modeled in two distinct phases: pull-down and continuous cooling. Usually, the trailer would be pulled down to the desired temperature before the pre-cooled goods are loaded into it. A drive-cycle, where these phases are visible, is shown in Fig. 5. Around one hour into the drive, the trailer has reached its set-point temperature, the pull-down phase ends, and the continuous cooling phase begins. While during the pull-down phase, the TRU cools with its maximum power, like in the cycle sentry mode, during the continuous mode, the TRU has an almost constant power consumption independent of the inside and ambient temperature. The trailer used in the test drives has a constant consumption of ≈ 6 kW during this continuous cooling mode, which was found experimentally, as shown in Fig. 5. This power consumption, however, is not directly translatable into a heat flow, because a controller controls the amount of cooled air blown into the trailer to keep the temperature constant while the access air is expelled.

This behavior can not be easily modeled, and doing so has no benefit for modeling drive cycles, because the energy consumption during continuous cooling is almost constant. Thus, we suggest splitting continuous driving cycles into two parts. While the trailer cools down, it can be modeled as a cycle-sentry drive. After the inside temperature has reached the desired set-point, power consumption can be assumed to be constant. For this reason, the rest of this paper focuses on modeling and parameter identification of the cycle-sentry operation mode. The fundamentals of the proposed parameter identification are described in the following Section.

3. Parameter identification

In addition to the general model for the refrigerated trailer, this paper also presents a novel method for parameter identification of the trailer's and good's parameters.

The parameters themselves can be split up into two specific groups: the trailer-specific parameters and the goods-specific parameters. The trailer-specific parameters can be estimated with all drive cycles, because they stay constant and comprise the specific heat capacity C_T of the trailer, the heat transfer coefficient K_T from the trailer to the outside, as well as the PLMs for the COP and TRU power consumption.

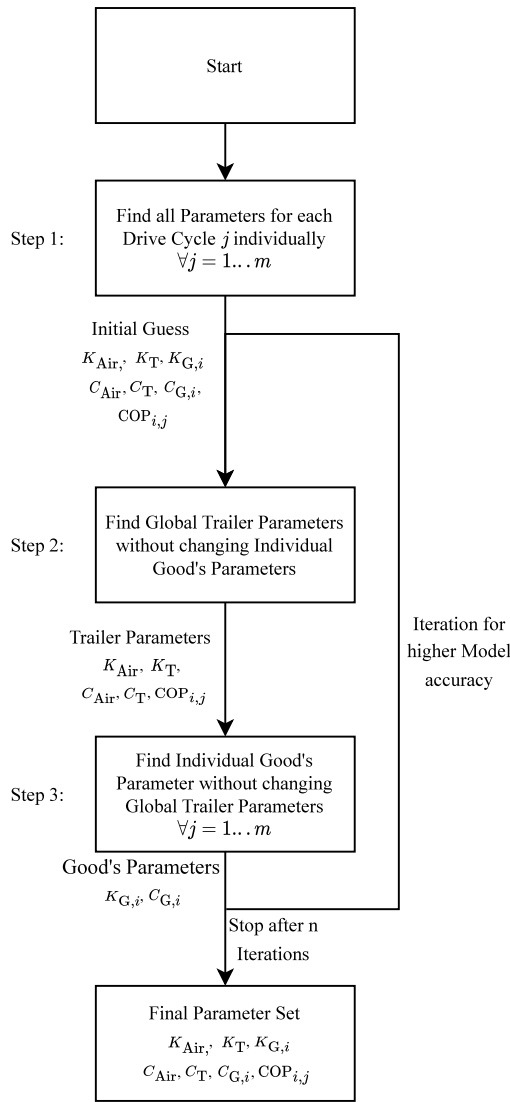


Fig. 6. Parameter identification process. First, initial parameters are found for each drive individually. Afterward, the trailer parameters are found using all drive cycles for evaluation. Then, each drive cycle's goods-specific parameters are optimized. Steps two and three can be repeated until the result converges or after a specified number of iterations were performed.

In contrast, because the loaded goods change between drives, the goods-specific parameters differ for each drive cycle. They consist of the individual heat capacities $C_{G,i}$ with which the goods are modeled, their heat transfer coefficients $K_{G,i}$, and initial temperatures $T_{G,0}$.

The identification process consists of three distinct steps. First, model parameters are found for each drive individually, including the trailer and the goods-specific parameters, with the Root Mean Square Error (RMSE) between the measured temperature $T_{Air,k}$ and simulated temperature $\hat{T}_{Air,k}$ for all time steps $k = 1 \dots n$ as the loss function

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (T_{Air,k} - \hat{T}_{Air,k})^2}. \quad (6)$$

The found initial trailer parameters are used as values to identify the global trailer parameters for all m drive cycles. This happens by changing the loss function to be the average root mean square error over all drive cycles while all cycles use the same trailer parameters.

$$Loss = \frac{1}{m} \sum_{j=1}^m RMSE_j \quad (7)$$

The good's parameters remain drive-specific and do not change during this step. Lastly, with the found global trailer parameters, each drive's good's parameters are re-identified; during this step, the trailer parameters stay constant. Steps two and three can be repeated multiple times until the result converges in order to ensure the best overall parameter set. The parameter identification process is also shown in Fig. 6.

As the parameters of the goods need to be identified individually for each drive cycle, the time spent in step three would scale linearly with the number of test drives. To reduce the computational resources needed, a maximum time limit was set for all drives, and time for identifying each cycle's parameters was allocated in correlation to the drive's error. Thus, drives with large errors had more time to identify the parameters, while drives with relatively small errors were allocated less time.

The parameters themselves were identified using a Particle Swarm Optimization (PSO). PSO is a computational method that optimizes a problem by iteratively trying to improve candidate solutions, regarded as particles in the search space. Each particle has a position x , velocity v , and a fitness value. The position of a particle represents a potential solution within the problem space. The velocity indicates both the magnitude and direction of the particle's movement in the search space. The fitness value measures the solution quality of the particle's current position (Kennedy and Eberhart, 1995). PSO was chosen as the parameter identification method over other genetic or gradient based algorithms, as the PSO generally converges for global non convex optimization problems faster and requires less computational resources (Almufti et al., 2019).

The position of each particle is updated according to the following equation:

$$x_i(k+1) = x_i(k) + v_i(k+1). \quad (8)$$

The velocity of each particle is updated using the subsequent equation:

$$v_i(k+1) = wv_i(k) + c_1r(pb_i - x_i(k)) + c_2r(gb - x_i(k)). \quad (9)$$

In this equation, w represents the particle's inertia, which controls the particle's tendency to continue in its current direction. The acceleration coefficients, c_1 , and c_2 determine how much a particle is drawn to its personal best position (pb) and the globally best position (gb) found by the swarm. The term r is a random number between 0 and 1, which introduces a stochastic element to the particle's movement.

The parameters were identified using 100 particles over 1000 iterations for each identification process. The bounds of the COP were set to the theoretical performance limits, the bounds for the heat capacities and heat transfer coefficients are given in Table A.4.

4. Test vehicle and dataset

In order to confirm the accuracy of the proposed model and identification process across a range of different operating conditions, a total of ten test drives were carried out. These test drives resulted in over 40 h of driving data collected on public roads. The drives took place in Germany to assess moderate temperatures, in Spain to evaluate hot temperatures, and in Sweden to test low ambient temperatures with varying loads and temperature set-points.

A European-style truck was used as the test vehicle together with a trailer that was equipped with a 40 kWh battery and a generator axle for braking energy recovery to power the electric TRU. The truck is shown in Fig. 7 and the main inside dimensions of the trailer are provided in Table 2. It was also equipped with two additional calibrated K-type thermocouple temperature sensors, with an error of ± 0.2 °C, to measure the inside and ambient temperatures. The inner thermocouple sensor can be seen in Fig. 8, which shows the loading area of the trailer. The TRU power consumption was measured directly using the inverter that was powering it, with a rated accuracy of $\pm 5\%$.



Fig. 7. The test vehicle used in this study is equipped with an electrically powered TRU, a 40 kWh battery, and a generator axle for braking energy recovery.

Table 2

Main inner dimensions and TRU parameters of the trailer.

Height in m	2.65
Length in m	13.3
Width in m	2.6
Volume in m ³	91.6
Air flow in m ³ /h	5000
Rated power of motor in kW	9.3
Rated $\dot{Q}_{TRU}$ @ 0 °C in kW	11.9
Rated $\dot{Q}_{TRU}$ @ -20 °C in kW	6.2



Fig. 8. The loading area of the trailer was filled with steel boxes, which were partly filled with concrete to simulate the transport of goods. The temperature inside is recorded using a thermocouple, which can be seen in the middle of the image, held up by the white wire.

Table 3

Summary of the ten different test drives conducted in Germany (Ger), Spain (Spa), and Sweden (Swe). A total of over 40 h of drive data on public roads was recorded.

Nr.	Ins. Temp. °C (min; max)	Amb. Temp. °C (min; max)	Length in hours	Region
1	-21.5; 15.5	11.2; 19.5	6.5	Ger
2	-19.3; -2.4	10.5; 16.9	2.8	Ger
3	-0.8; 18.6	14.9; 21.4	1.0	Ger
4	-16.0; 1.0	10.2; 16.9	3.3	Ger
5	-19.7; 8.3	12.6; 24.9	5.6	Ger
6	-1.5; 15.8	12.4; 27.2	7.6	Ger
7	-23.1; -3.9	0.8; 5.1	4.1	Swe
8	-20.5; -15.8	21.6; 35.9	5.4	Spa
9	-20.4; -16.3	26.8; 33.4	3.8	Spa
10	-1.3; 12.0	19.4; 24.2	1.0	Ger

To simulate the transported goods, the trailer was loaded with 18 steel boxes, each weighing 180 kg empty. Between the test drives, a varying number of these steel containers were filled with 320 kg of concrete, resulting in a total weight of 500 kg for a filled container. The test drives were conducted with both pre-cooled and ambient temperature goods to further show the robustness of the proposed model for varying load conditions.

An overview of all conducted drive cycles, their minimum and maximum inside and ambient temperatures, as well as their length, is given in Table 3.

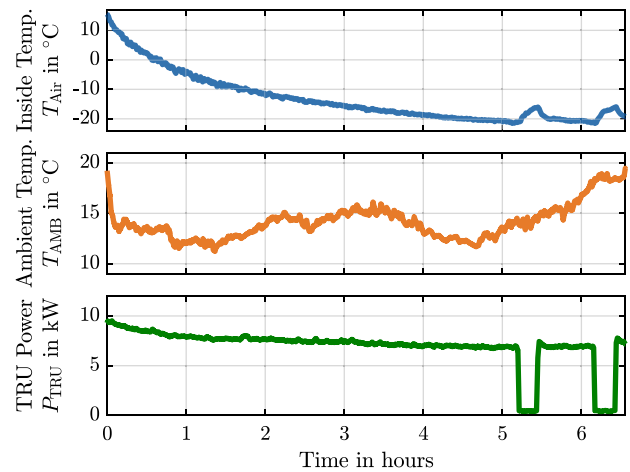


Fig. 9. Drive cycle number 1 conducted in Germany. It has a low set-point of -18 °C, leading to the initial cooling down of the trailer and goods taking around five hours, as the goods were not pre-cooled. The influence of the inside temperature on the TRU power consumption can be seen as the true power drop combined with the inside temperature.

Four of the ten test drives will be discussed in more detail in the following, because they represent the lower and upper limits of the recorded ambient and inside temperatures; the remaining six can be found in the Appendix.

Test drive number one, conducted in Germany, is shown in Fig. 9; it starts within a building, leading to the sharp drop in the ambient temperature at the beginning of the drive. The inside temperature is around 15 °C at the start of the cycle and gets pulled down to -21.5 °C, while the ambient temperature ranges between 11 °C and 20 °C. In this drive cycle, it is especially noticeable that the TRU power consumption can not be assumed to be constant, as it drops from almost 10 kW to under 7 kW with the falling inside temperature of the trailer, showing the need to model the power consumption of the TRU in dependence of the inside and ambient temperatures. As the trailer and goods have an initial temperature of 15 °C at the beginning of the drive, the trailer takes more than 5 h to reach the set-point of -18 °C for the first time and only completes two on/off cycles of the TRU.

Drive cycle number 6, which was recorded in Germany as well, is shown in Fig. 10. The dynamics of the cycle sentry mode can be seen in it. During the 7-h drive, the TRU switches off and on a total of nine times after an initial pull-down phase of approximately 45 min.

Fig. 11 depicts a test drive taken in Sweden in order to include a test drive with a low ambient temperature, ranging between 1 °C and 5 °C. The trailer starts pre-chilled and is cooled down further during the four-hour drive.

Lastly, Fig. 12 depicts one of the test drives conducted in Spain in order to have test drives with high ambient temperatures, which range between 21 °C and 36 °C. The trailer and goods were pre-cooled to -20 °C. In this cycle, the influence of the outside temperature on the TRU consumption is visible, because it rises with the increasing outside temperatures despite similar inside temperatures. Furthermore, it is visible after hour two that due to the elevated ambient temperatures, the TRU needs to be powered on for much longer to reach the lower temperature band compared to the first two hours.

After explaining the fundamental ideas of the model and parameter identification approach and discussing the test drives, the following chapters will focus on validating the model and its accuracy, starting with open-loop temperature modeling.

5. Results of open-loop temperature modeling

To validate the accuracy of the thermal box model of the trailer and the model of the transported goods, an open-loop temperature simulation was conducted. It uses the measured ambient temperature as well

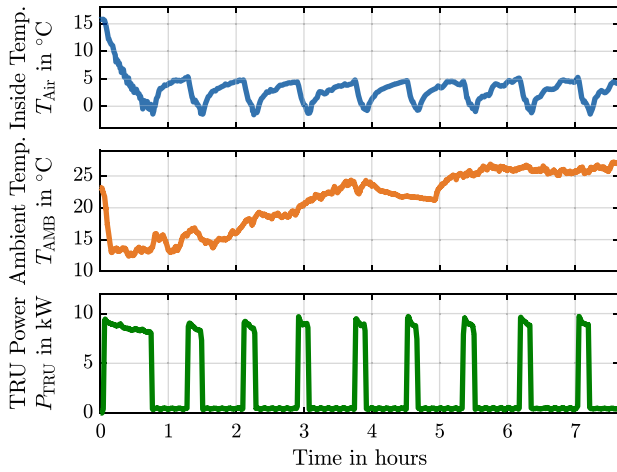


Fig. 10. Drive Cycle 6 recorded in Germany shows a drive cycle, which symbolizes the transport of chilled goods, with the temperature band ranging between 0 °C and 5 °C, containing a total of 9 on/off-cycles of the TRU.

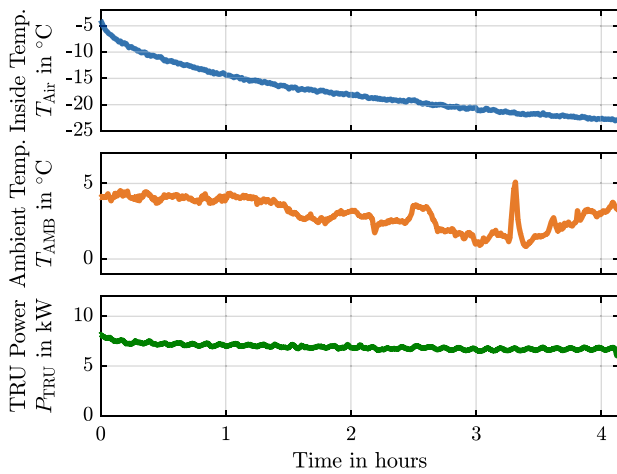


Fig. 11. Drive cycle 7 conducted in Sweden is an example of a drive cycle with a low ambient temperature ranging between 1 °C and 5 °C. The inside temperature drops because the trailer does not reach a thermal equilibrium or its set-point of -23 °C during the drive cycle.

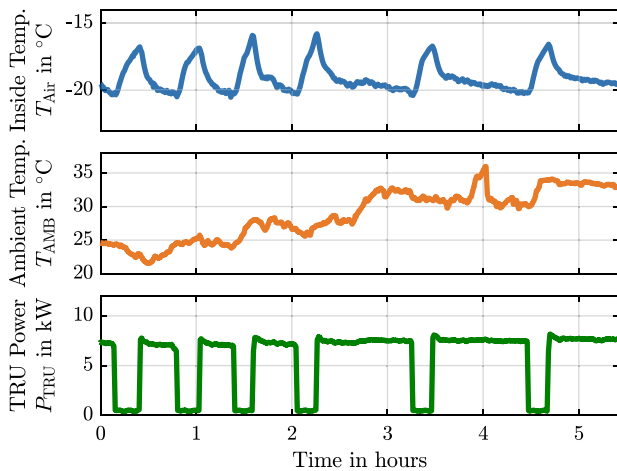


Fig. 12. Drive cycle 8 was driven in Spain with hot outside temperatures, reaching over 35 °C and a low set-point of -20 °C.

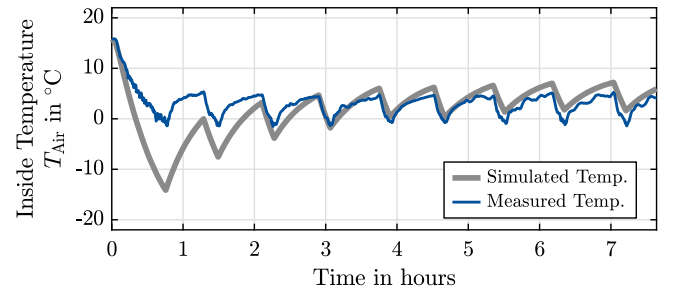


Fig. 13. Example of misidentified parameters. Due to the open-loop nature of the temperature simulation, errors can accumulate. The model has no knowledge of the temperature set-point or the current measured inside temperature.

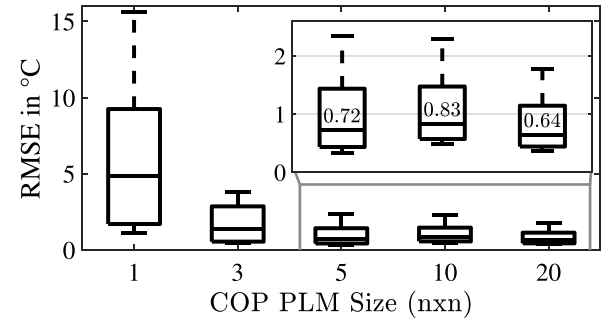


Fig. 14. Boxplot of the RMSE for the validation half of each of the ten test drives. Already, a 5×5 PLM is sufficient for a good model quality, with a median RMSE of 0.72 °C.

as the measured TRU power as inputs and models the inner temperature of the trailer without using the modeled inside temperature as feedback for the controller. To achieve this, the parameters for the trailer and goods are identified during the first half of each test drive and then validated in the second half. The model takes into account the outside temperature and the measured TRU power consumption, along with the initial temperatures for the goods and the trailer. However, it does not have information about the temperature set-point. Additionally, due to the open-loop simulation, even small errors accumulate over time, as is shown exemplarily in Fig. 13.

5.1. Comparison of different COP PLM sizes

As described in Section 2, the COP is modeled using a PLM. In this section, the required size for the PLM will be determined. To do so, the goods are modeled as two heat capacities, and the trailer is modeled with COP PLMs of the sizes ($n \times n$) 1, 3, 5, 10, and 20, which use the current modeled inside air temperature $\hat{T}_{Air,k}$ and measured ambient temperature $T_{Amb,k}$ as input (see Fig. 14).

Already, a 5×5 PLM shows good agreement with a median Root Mean Square Error (RMSE) of 0.72 °C over all drives. The best-performing configuration was a 20×20 PLM with an RMSE of 0.64 °C. Modeling the COP as constant, on the other hand, resulted in much worse prediction accuracy, showing that due to the wide variety of operating conditions in which a refrigerated trailer must perform, it is not possible to assume a constant COP value.

While a RMSE of 0.7 °C is higher than the maximum error of 0.3 °C reported by Artuso et al. (2019), this can be explained, because their model parameters were identified on the whole test, i.e., the 0.3 °C refer to the error on the identification data and not on separate test data. Furthermore, the results presented here were obtained with test drives under varying conditions and not in a controlled test setup. Sørensen et al. report an error of up to 0.8 °C (Sørensen et al., 2015a), comparable to our results, with one set of goods varying the cooling

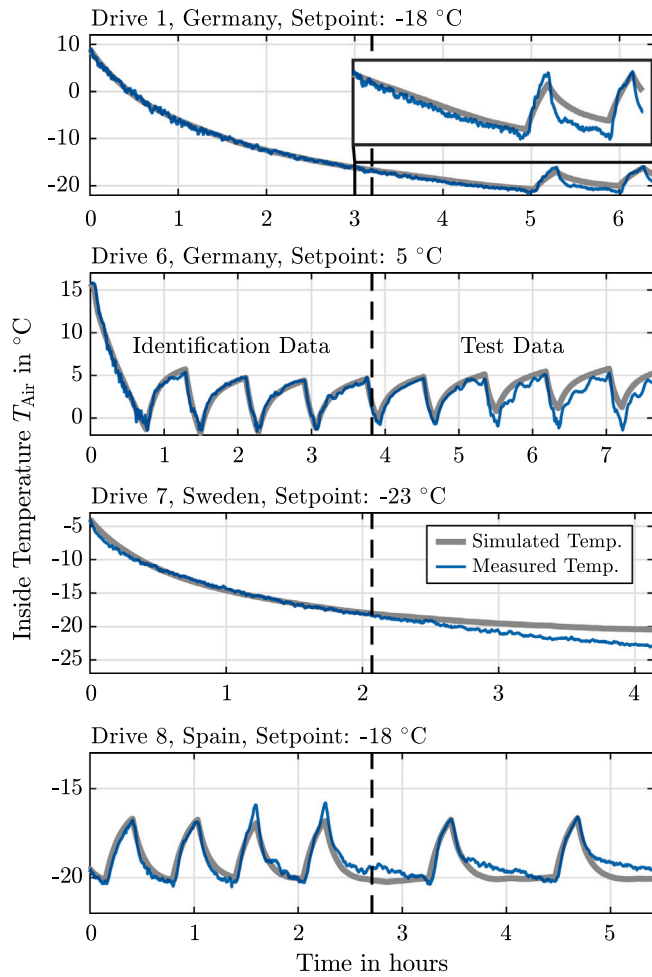


Fig. 15. Results of the open loop temperature simulation, where the first half of each test drive is used for parameter identification and the second half for validation, shows very high agreement between the simulated and measured temperature in dynamic runs.

capacity. The proposed model performs similarly despite varying loads, temperature set-points, and ambient temperatures between the runs, which is promising.

The simulation results for the four previously discussed test drives using a 5×5 PLM are given in Fig. 15.

The model and measurements show good agreement, even after multiple hours on the never-before-seen second half of the test drives, despite the simulation being open-loop. However, it is noticeable that the parameters of drives with greater temperature dynamics, such as the second and fourth drives in Fig. 15, are identified more accurately, leading to a better fit. In drive seven a higher error is noticeable, as the identification time is shorter and the temperature dynamic during it is smaller, leading to a larger drift from the measured temperature over time. This suggests that in drives with short identification times or low temperature dynamics, the model can overfit leading to higher errors than in drives with a large dynamic.

5.2. Goods model evaluation

After identifying that a 5×5 PLM is sufficient to model the COP with a high agreement between model and measurement using a two-capacity goods model, this section will focus on the number of heat capacities needed to accurately model the goods. The previously identified 5×5 PLM is employed to test the accuracy of the different goods models.

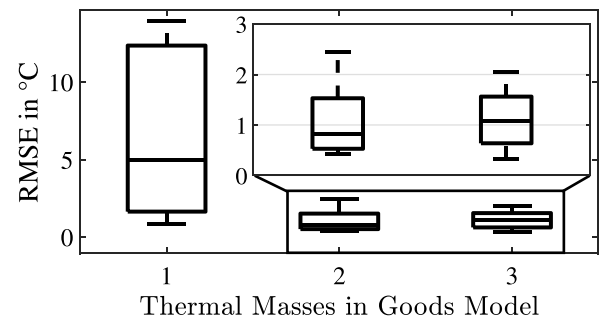


Fig. 16. Boxplot of the RMSE for the one, two, and three heat capacity goods models for all ten test drives.

The results of all drives are shown in Fig. 16.

Modeling the goods as a single thermal mass resulted in a large disagreement between the simulation and the measurements and a median RMSE of 5 °C over all runs. Modeling the goods as two or even three separate capacities yielded much better results, with 0.7 °C and 1.1 °C median RMSE, respectively.

In our tests, the commonly used assumption of modeling the goods as a single heat capacity performs an order of magnitude worse than the two or three-capacity models between which no more large improvements were noticeable.

6. TRU power consumption model

After identifying and validating the parameters for the thermal models of the goods and the trailer, the TRU power consumption PLM is identified and validated.

Fig. 17 shows the simulated and measured powers using a 3×3 PLM to estimate the power consumption, showing a high degree of agreement between the estimation and the measurement overall inside and outside temperature ranges. The largest disagreements occur when the TRU powers on, because the power demand spikes during the first seconds. The RMSEs for all drives and the different PLM sizes are shown in Fig. 18. The median RMSE only improves slightly despite the increasing PLM size and is around 0.13 kW for the 3×3 PLM, mainly caused by fluctuations of the measurement around the model as can be seen in Fig. 17, suggesting that a 3×3 PLM is sufficient to model the TRU power consumption.

7. Complete closed-loop model

Finally, the temperature and TRU power consumption models are combined to build a closed-loop model, in which the TRU ON/OFF-State is set based on the modeled air temperature $\hat{T}_{Air,k}$. The TRU controller is implemented as a bang-bang controller, keeping the inside temperature between the two temperature bands. The model's known inputs are the initial temperatures, the ambient temperature over the drive, and the upper and lower temperature limits of the TRU controller.

The example of drive cycle eight is shown in Fig. 19.

The closed-loop model drifts slightly from the real-world measurements; however, the total energy consumption of the simulated and real TRU differs by only 1.3% in drive cycle eight and under 1.5% on average for drive cycles 6, 8, and 9. The error between the model and the measured inside temperature results in the bang-bang controller triggering at different times in model and reality. This leads to a temporal drift that can potentially add up over time. Furthermore, the temperature sensor for measuring the inside temperature is positioned at a different position than the temperature sensor of the TRU; this can be seen, for example, at the third and fourth on/off cycles, where the

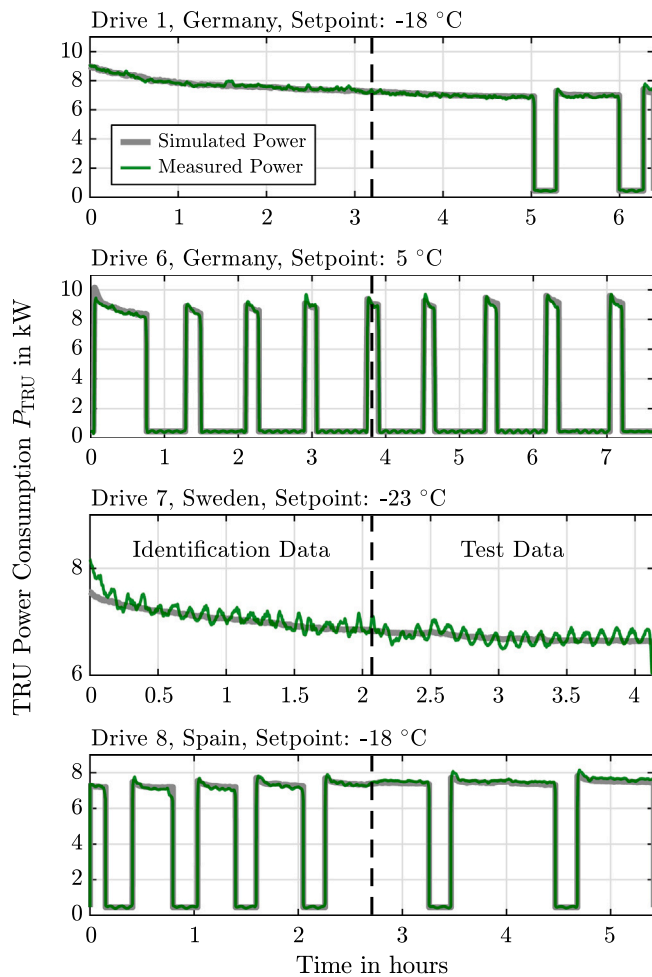


Fig. 17. Results of power consumption model using a 3×3 PLM showing a high agreement between the model and the measurement under a wide variety of inside and ambient temperatures.

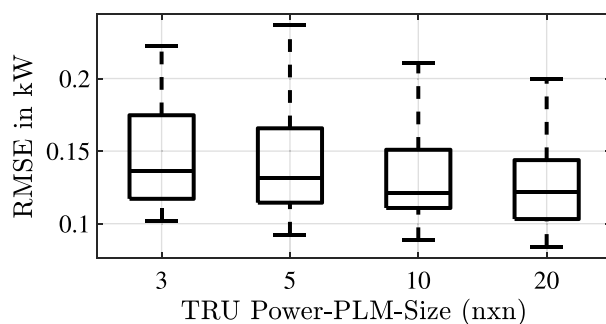


Fig. 18. Boxplot of the RMSE of the TRU over all drive cycles based on the used PLM size, already a 3×3 PLM has a median RMSE of 0.13 kW, increasing the PLM size improves the power consumption estimation only slightly.

TRU is triggered at a higher temperature than the set-point due to the uneven temperature distribution inside of the trailer.

The small error of total energy consumption, even over different drive cycles with various loads and set-points, makes the model suitable for estimating the TRU power consumption over a complete drive cycle. Thus, it can be used for either drive cycle simulation or as a base for a model predictive control.

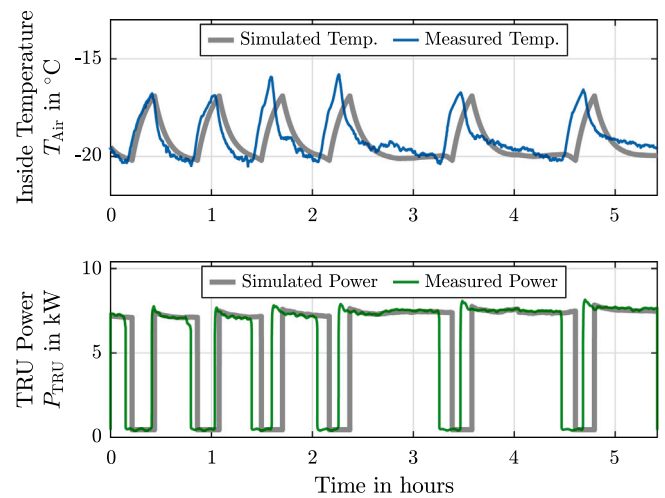


Fig. 19. Drive cycle 8 recorded in Spain in a closed-loop simulation with only the start and ambient temperatures as inputs. The error in total TRU power is around 1.3%.

8. Conclusion and outlook

An accurate model of a refrigerated trailer is essential for drive cycle simulation and developing MPC-based control strategies to reduce the CO₂ emissions associated with refrigerated transport. This paper presented a novel modeling and parameter identification approach for refrigerated trailers and transported goods. The proposed model combines the solar influx and the air-leakage as part of the heat transfer coefficient of the trailer, thus only requiring the ambient temperature as well as the set-point temperature and an estimation of the goods' parameters as inputs. This enables the model to be used in future works for an MPC based control in systems, for example, to estimate the total energy needed for the drive to reduce the energy supplemented by a combustion engine to the required minimum. To further improve the practical applicability, the model can be expanded in future works to include the detailed modeling of the air infiltration during trailer door openings, which would be especially relevant in urban delivery cycles, where many door openings occur. Another relevant future expansion of the model is the heat generated by goods due to respiration, especially in transporting fresh goods, as it increases the energy required to cool down the trailer.

These are the main results of this study:

- A new thermal trailer model was proposed that does not need the solar influx and air leakage as inputs and models the loaded goods as multiple heat capacities.
- A novel three-step parameter identification process was presented, in which the trailer and good's parameters are identified separately, using the measured inner and ambient temperature and the TRU power as inputs, allowing for the first time to use multiple different drives, with unknown loads, to identify the trailer's parameters.
- The findings were validated with over 40 h of driving data in three countries, representing the most diverse data set used so far to validate trailer models.
- The thermal model performs comparable to the state of the art despite being validated under a broader range of temperatures, demonstrating the model's robustness. It was found that modeling the goods as two heat capacities drastically improves the model's accuracy for goods that are not pre-cooled from 4.9 °C to 0.7 °C.
- This study presents, to our knowledge, the first validated power consumption model of an electric TRU and the corresponding COP, with which the total energy consumption over a drive can be modeled with less than 1.5% error.

Table A.4

Lower and upper bounds for the heat capacities and heat transfer coefficients used for the PSO.

Parameter	Bounds [Lower Upper]
C_{Air}	[1e4 1e5]
C_T	[1e3 1e10]
$C_{G,i}$	[1e2 1e8]
K_{Air}	[5e2 5e4]
K_T	[50 500]
$K_{G,i}$	[1e2 1Se8]

- The proposed model and parameter identification process leads to a more robust trailer model, which, among other things, could be used for an MPC of electrified refrigerated trailers, reducing their power consumption while ensuring that the cooling of the goods is uninterrupted.

Further research might look into:

- The expansion of the goods model to include respiratory heat, which is generated by the sugars in the food breaking down and thus releasing heat.
- The modeling of air infiltration during door openings, the explicit modeling of the solar influx, the expansion of the trailer and TRU model in order to be able to simulate multiple temperature zones and temperature distributions within one trailer.
- A sensitivity analysis determining the influence of humidity, solar influx, wind speed and other environmental factors on the performance of the model
- A direct comparison with different trailer models especially CFD approaches
- Utilization of the proposed model in an MPC in order to further optimize refrigerated trailers with more than one energy source, such as hydrogen electric trailers or trailers that have a generator axle.
- The fast identification of the good's parameters with known trailer parameters as it would allow a potential MPC to react quickly to changes in the loaded goods.

CRediT authorship contribution statement

D. Bank: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J.P. Kobler:** Data curation. **T. Zeller:** Writing – review & editing. **P. Cujic:** Writing – review & editing. **T. Seel:** Writing – review & editing, Writing – original draft, Conceptualization. **S.F.G. Ehlers:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the project manager “German Aerospace Center” (DLR), for managing the project “RekuTrAx” (project number 01MV22018) and the “Federal Ministry for Economic Affairs and Climate Action of Germany” (BMWK) for the funding of this research.

Appendix

See Figs. A.20–A.25 and Table A.4.

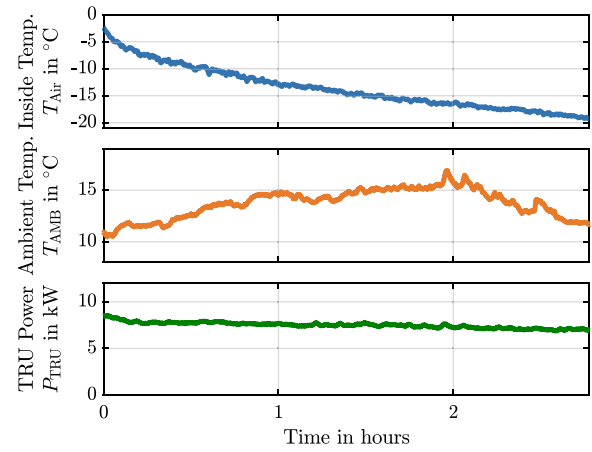


Fig. A.20. Drive Cycle 2 conducted in Germany.

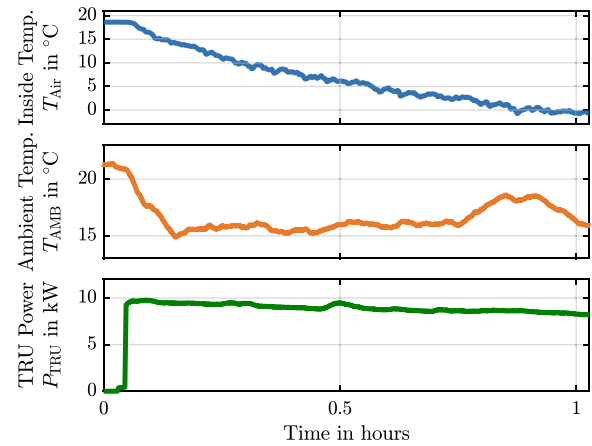


Fig. A.21. Drive Cycle 3 conducted in Germany.

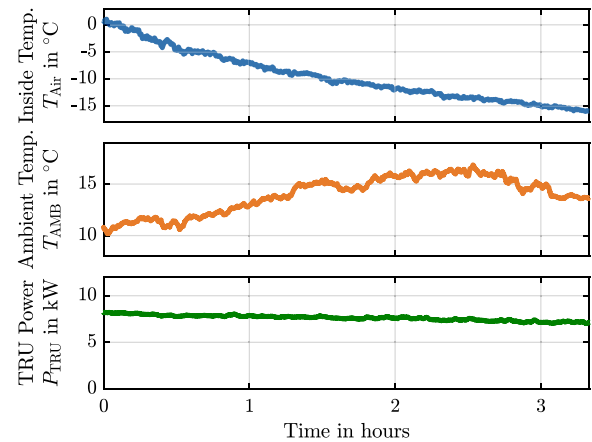


Fig. A.22. Drive Cycle 4 conducted in Germany.

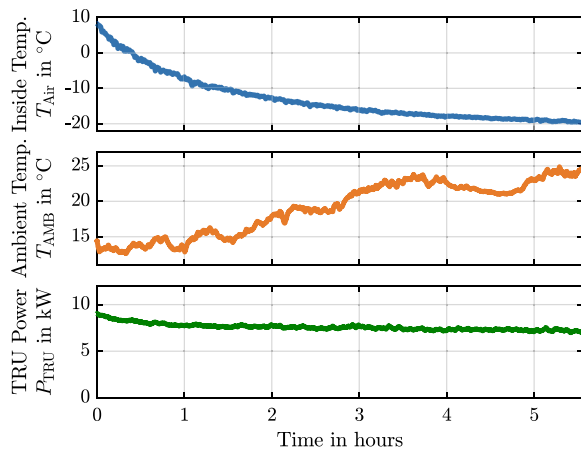


Fig. A.23. Drive Cycle 5 conducted in Germany.

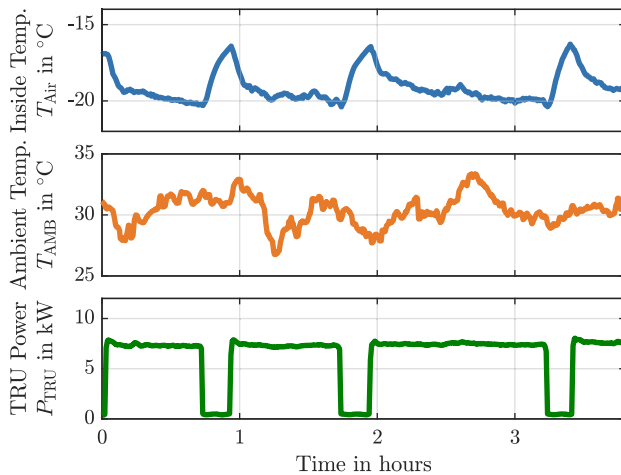


Fig. A.24. Drive Cycle 9 conducted in Spain.

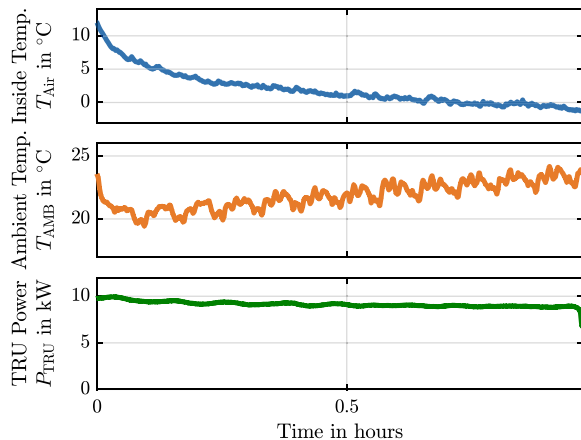


Fig. A.25. Drive cycle 10 conducted in Germany.

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